

Nikko J. Cleri

Eberly Postdoctoral Fellow
The Pennsylvania State University

Summary

Research:	Galaxy Evolution, High-Redshift Galaxies, Emission-Line Galaxies, Interstellar Medium, Active Galactic Nuclei, Black Hole Seeds, Population III Stars, Star Formation, Dust Attenuation
Techniques:	UV/Optical/Near-IR Spectroscopy, Photoionization Modeling
Proposals:	As PI: 2 JWST, 1 HST, 29.2 JWST hours and >0.5M USD awarded As Co-I: 14 JWST, 1 Gemini, 1 HET, >850 total hours awarded
Publications:	8 as lead/co-lead author, 104 total, 7679 citations, 43 h-index
Presentations:	23 research, 17 outreach and professional development

Academic and Professional Appointments

2024-	Eberly Postdoctoral Fellow	Penn State
2021-24	Graduate Student (Ph.D.)	Texas A&M
2019-21	Graduate Student (M.S.)	UConn
2017-20	Research Assistant (B.S.)	UConn
2018	NSF REU Student	Texas A&M

Education

Ph.D. Astronomy	Texas A&M University	2021 - 2024
Advisor: Casey Papovich		
Thesis: <i>Spectroscopic Studies of Stars and Black Holes Across Cosmic Time</i>		
M.S. Physics	University of Connecticut	2019 - 2021
Advisor: Jonathan R. Trump		
Thesis: <i>CLEAR: Paschen-β Star Formation Rates and Dust Attenuation in Low Redshift Galaxies</i>		
B.S. Physics Mathematics Minor	University of Connecticut	2015 - 2019
Advisor: Gerald V. Dunne		
Undergraduate Research: <i>Resurgent trans-series for generalized Hastings-McLeod solutions</i>		

Awarded Proposals and Grants

Principal Investigator		3
2026	JWST-GO-10518: <i>CEDAR: Characterizing Electron Densities At Reionization</i>	29.2 hours/\$234k
2024	JWST-AR-5558: <i>A Census of Optical Diagnostics of Ionizing Sources Across Cosmic Time</i>	\$174k
2021	HST-AR-16609: <i>Peering Through the Dust: Paschen-beta Indicators of Star Formation and Dust Attenuation</i>	\$136k

Co-Investigator		16
2026	JWST-GO-11376: <i>Water in Little Red Dots? A Pathfinder Program with the MIRI/LRS to Confirm the Cool Temperatures of LRDs</i> (PI: B. Wang, J. Helton)	13.0 hours
2026	JWST-GO-10445: <i>Reddened by dust or gas? Unveiling the "Red" in Little Red Dots with MIRI/LRS</i> (PI: A. de Graaff, D. Setton)	49.1 hours
2026	JWST-GO-10311: <i>Slicing the Saguaro: Dissecting a Little Red Dot Descendant at Cosmic Noon with NIRSpec/IFU</i> (PI: P. Rinaldi, E. Iani)	26.7 hours
2025	HET-PSU26-1-021: <i>The Hobby-Eberly Telescope Void Galaxy Sample</i> (PI: O. Curtis)	26 hours
2025	JWST-GO-8047: <i>Extremely massive galaxies in the early universe? Confirming the nature of the most model-breaking object by hunting for stellar absorption features</i> (PI: B. Wang and E. Nelson)	19.6 hours
2025	JWST-GO-7488: <i>Echoes of Silence: Absorption Line Spectroscopy of a Massive Quiescent Galaxy at $z=7.3$</i> (PI: A. Weibel)	14.1 hours
2025	JWST-GO-8559: <i>SPAM: Star-formation from Photometry through the Addition of Medium-bands</i> (PI: K. Davis and R. Larson)	62.8 hours
2025	JWST-GO-8410: <i>A Census of Galaxy Kinematics and Outflows to $z \sim 7$</i> (PI: R. Simons)	110.0 hours
2025	JWST-GO-8204: <i>Give me a break: the search for stars in a prototypical Little Red Dot</i> (PI: J. Greene and I. Labbe)	17.4 hours
2024	JWST-GO-5718: <i>A Spectroscopic Census of Faint, Broad-Line AGN at $z>5$</i> (PI: D. Kocevski and J. Guo)	20.49 hours
2024	JWST-GO-5943: <i>What really are the Physical Properties of Galaxies in the Epoch of Reionization?</i> (PI: C. Papovich and W. Hu, T. Hutchison)	61.83 hours
2024	JWST-GO-5407: <i>MEOW: The MIRI Early Obscured-AGN Wide Survey</i> (PI: G. Leung and R. Endsley, S. Finkelstein)	73.95 hours
2024	JWST-GO-5507: <i>Deep Spectroscopy of Galaxies at $z=4-14$: Uncovering Drivers of Early Galaxy Formation and Black Hole Growth</i> (PI: T. Hutchison and R. Larson)	23.29 hours
2024	JWST-GO-6368: <i>The CANDELS-Area Prism Epoch of Reionization Survey (CAPERS)</i> (PI: M. Dickinson)	293.21 hours
2023	JWST-GO-3703: <i>Breaking the $z=10$ barrier with MIRI: redshift confirmation and detection of rest-frame optical emission lines</i> (PI: J. Zavala)	24.33 hours
2023	GS-2023A-Q-136: <i>Optical Spectroscopy of JWST ERO Galaxies</i> (PI: B. Backhaus)	20 hours

Honors and Awards

2024	Dean's Climate and Diversity Award (Group)	Penn State
2022	Texas Space Grant Consortium Graduate Fellowship - \$5K	Texas A&M
2018	NSF REU - \$5K	Texas A&M
2016	Dean's List - College of Liberal Arts and Sciences	UConn
2015-19	Governor's Scholarship - \$8.5K/yr	UConn
2015	Community Service Scholarship - \$1K	UConn

Teaching Experience

Instructor of Record

ASTRO 6: Stars, Galaxies, and the Universe	Penn State
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Guest Lecturer

ASTRO 502: Fundamental Astrophysics	Penn State
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Teaching Assistant/Course Assistant

PHYS 1501: Physics for Engineers I	UConn
PHYS 1025: Introduction to Astronomy	UConn

Service

Physics & Astronomy Community

Organizing Committee - 8th IGC Neighborhood Workshop, State College, PA, April 2026

Referee – Astronomy & Astrophysics (A&A)

Referee – Astrophysical Journal (ApJ)

College

Astronomy Representative – Eberly College of Science Postdoctoral Council	Penn State
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Department

Founder/Organizer – ExGal Journal Club	Penn State
Member – Climate and Diversity Committee	Penn State
Graduate Representative	Texas A&M
Organizer – Astronomy Journal Club	Texas A&M

Mentoring

Students Supervised

Primary mentor for 3 undergraduates at Penn State	2024-
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Other Mentoring Activities

Co-Founder – Preparing the Emerging Next Generation Using Inclusive Networking (PENGUIN)	
Primary Mentor – Penn State-Nanjing Exchange Program	Penn State
Coordinator – Mentoring and Advising Graduates in an Inclusive Community (MAGIC)	Texas A&M
Mentor – Mentoring and Advising Graduates in an Inclusive Community (MAGIC)	Texas A&M
Mentor – UConn Undergraduate Peer Mentoring	UConn

Outreach

Presenter – Astronomy on Tap State College	Penn State
Volunteer – Gateway to Graduate School	Texas A&M

Demonstrator – Physics and Engineering Festival	Texas A&M
High School Research Reviewer – Lumiere	Texas A&M
Presenter – Astronomy on Tap BCS ‘In the News’	Texas A&M
Treasurer – Astronomy on Tap BCS	Texas A&M
Pen-Pal – Letters to a Pre-Scientist	Texas A&M
Volunteer – Mitchell Institute Star Party Group	Texas A&M

Press

A Week Downeast, Challenging Assumptions about Galaxy Evolution, Astrobites, O. Cooper et al. 2025

Collaborations and Survey Membership

Principal Investigator

CEDAR: Characterizing Electron Densities At Reionization	JWST
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Co-Investigator

SPAM: Star-formation from Photometry through the Addition of Medium-bands	JWST
CAPERS: The CANDELS-Area Prism Epoch of Reionization Survey	JWST
THRILS: The High [Redshift+Ionization] Line Search	JWST

Contributing Member

LEGGOS: Lensing and Galaxy Growth: Observing Substructures	JWST
POPPIES: The Public Observation Pure Parallel Infrared Emission-Line Survey	JWST
RUBIES: Red Unknowns: Bright Infrared Extragalactic Survey	JWST
NGDEEP: The Next Generation Deep Exploratory Public Survey	JWST
CEERS: The Cosmic Evolution Early Release Science Survey	JWST
CLEAR: The CANDELS Ly α Emission at Reionization Survey (inactive)	HST

Technical Skills and Programming Languages

Languages	Fluent: Python, LaTeX
	Familiar: SQL, Julia, C, C++, R, IDL, perl, Mathematica, MATLAB, HTML, CSS
Software	Fluent: Cloudy, PyNeb, LiMe
	Familiar: grizli, DS9, IRAF, sbatch, slurm

Website Architect

Personal Website: njcleri.github.io
Mentoring and Advising Graduates in an Inclusive Community (MAGIC) (co-author): tx.ag/tamumagic
Preparing the Emerging Next Generation Using Inclusive Networking (PENGUIN): penguin-mentoring.github.io

Publications

A current list of my papers can be found on SciX here.

Lead/Co-Lead/Student-Led Papers

† = Undergraduate-led papers

8. **Cleri, N. J.**, et al. 2026, *RUBIES: The Evolution of the Ionization Parameter from $0 < z < 9$* , arXiv e-prints, arXiv:2605.30410 (submitted to ApJL)
7. †Ge, S.-R., **Cleri, N. J.**, et al. 2025, *JWST/NIRSpec Reveals a Small Population of Dominant Dust-Obscured Ionizing Sources in Galaxies at $1 < z < 3$* , arXiv e-prints, arXiv:2511.08671 (submitted to ApJ)
6. **Cleri, N. J.**, Olivier, G. M., Backhaus, B. E., et al. 2025, *Optical Strong Line Ratios Cannot Distinguish Between Stellar Populations and Accreting Black Holes at High Ionization Parameters and Low Metallicities*, ApJ, 994, 146,
5. Backhaus, B. E., **Cleri, N. J.**, et al. 2025, *Emission-Line Diagnostics at $z > 4$: $[O III]\lambda 4364/H\gamma$ versus $[Ne III]\lambda 3870/[O II]\lambda 3730$* , ApJ, 994, 125
4. **Cleri, N. J.**, Olivier, G. M., Hutchison T. A., et al. 2023, *Using $[Ne V]/[Ne III]$ to Understand the Nature of Extreme-Ionization Galaxies*, ApJ, 953, 10
3. **Cleri, N. J.**, Yang, G., Papovich, C., et al. 2023, *CLEAR: High-Ionization $[Ne V] \lambda 3426$ Emission-line Galaxies at $1.4 < z < 2.3$* , ApJ, 948, 112
2. **Cleri, N. J.**, Trump, J. R., Backhaus, B. E., et al. 2022, *CLEAR: Paschen- β Star Formation Rates and Dust Attenuation of Low Redshift Galaxies*, ApJ, 929, 3
1. **Cleri, N. J.**, Dunne, G. V., 2020, *Resurgent trans-series for generalized Hastings-McLeod solutions*, Journal of Physics A: Mathematical General, 53, 355203

Significant Author

4. Larson, R. L., Finkelstein, S. L., Kocevski, D. D., Hutchison, T. A., Trump, J. R., Arrabal Haro, P., Bromm, V., **Cleri, N. J.**, et al. 2023, *A CEERS Discovery of an Accreting Supermassive Black Hole 570 Myr after the Big Bang: Identifying a Progenitor of Massive $z > 6$ Quasars*, ApJL, 953, L29
3. Backhaus, B. E., Bridge J. S., Trump, J. R., **Cleri, N. J.**, et al. 2023, *CLEAR: Detecting Low-Luminosity Active Galactic Nuclei at $0.6 < z < 1.3$ via Spatially Resolved Hubble Space Telescope Grism Emission Line Ratios*, ApJ, 943, 37
2. Prescott, M. K. M., Finlator, K. M., **Cleri, N. J.**, et al. 2022, *Using Multiple Emission Line Ratios to Constrain the Slope of the Dust Attenuation Law*, ApJ, 928, 71
1. Backhaus, B. E., Trump, J. R., **Cleri, N. J.**, et al. 2022, *CLEAR: Emission Line Ratios at Cosmic High Noon*, ApJ, 926, 161

Co-Author: Published

77. Helton, J. M., et al. 2025, *Ionizing Photon Production Efficiencies and Chemical Abundances at Cosmic Dawn Revealed by Ultra-Deep Rest-Frame Optical Spectroscopy of JADES-GS-z14-0*, ApJL, 1004, L30
76. Brooks, M., et al. 2025b, *Beyond the Monsters: A More Complete Census of Black Hole Activity at Cosmic Dawn*, ApJ, 1002, 129
75. Arevalo-Gonzalez, F., et al. 2025, *The AGN nature of strong CIII] emitters in the Early Universe with JWST*, A&A, 709, A46
74. Wang, B., et al. 2025, *The Missing Hard Photons of Little Red Dots: Their Incident Ionizing Spectra Resemble Massive Stars*, ApJ, 1003, 10

73. Llerena, M., et al. 2025, *Extreme equivalent width-selected low-mass starbursts at $z = 4 - 9$: insights into their role in cosmic reionization*, A&A, 708, A152
72. Papovich, C., et al. 2026, *Galaxies in the Epoch of Reionization Are All Bark and No Bite – Plenty of Ionizing Photons, Low Escape Fractions*, ApJ, 1000, 111
71. Perry, M. N., et al. 2025, *The Prevalence of Bursty Star Formation in Low-Mass Galaxies at $z=1-7$ from $H\alpha$ -to-UV Diagnostics*, ApJ, 994, 14
70. Hviding, R. E., et al. 2026, *The X-Ray Dot: Exotic Dust or a Late-Stage Little Red Dot?*, ApJL, 1000, L18
69. Gandolfi, G., et al. 2026, *Mysteries of Capotauro - investigating the puzzling nature of an extreme F356W-dropout*, A&A, 706, A364,4
68. Zhang, Y., et al. 2026, *RUBIES Spectroscopically Confirms the High Number Density of Quiescent Galaxies from $2 < z < 5$* , ApJ, 997, 252
67. Setton, D. J., et al. 2025, *Little Red Dots at an Inflection Point: Ubiquitous “V-Shaped” Turnover Consistently Occurs at the Balmer Limit*, ApJ, 995, 118
66. Hviding, R. E., et al. 2025, *RUBIES: A Spectroscopic Census of Little Red Dots; All V-Shaped Point Sources Have Broad Lines*, A&A, 702, A57
65. Mascia, S., et al. 2025, *Little impact of mergers and galaxy morphology on the production and escape of ionizing photons in the early Universe*, A&A, 701, A122,
64. Leung, G. C. K., et al. 2024, *Exploring the Nature of Little Red Dots: Constraints on AGN and Stellar Contributions from PRIMER MIRI Imaging*, ApJ, 992, 26
63. Setton, D. J., et al. 2025, *A Confirmed Deficit of Hot and Cold Dust Emission in the Most Luminous Little Red Dots*, ApJL, 991, L10
62. Katz, H., et al. 2025, *21 Balmer Jump Street: The Nebular Continuum at High Redshift and Implications for the Bright Galaxy Problem, UV Continuum Slopes, and Early Stellar Populations*, The Open Journal of Astrophysics, 8, 104
61. Kokorev, V., et al. 2025, *CAPERS Observations of Two UV-Bright Galaxies at $z>10$. More Evidence for Bursting Star Formation in the Early Universe*, ApJL, 988, L10
60. Burgarella, D., et al. 2025, *CEERS: Possibly forging the first dust grains in the universe: A population of galaxies with spectroscopically derived extremely low dust attenuation (GELDA) at $4.0 < z < 11.4$* , A&A, 699, A336
59. Taylor, A. J., et al. 2025, *Broad-line AGNs at $3.5 < z < 6$: The Black Hole Mass Function and a Connection with Little Red Dots*, ApJ, 986, 165
58. Llerena, M. et al. 2025, *The ionizing photon production efficiency of star-forming galaxies at $z \sim 4 - 10$* , A&A, 698, A302
57. Brooks, M., et al. 2024, *Here There Be (Dusty) Monsters: High-redshift Active Galactic Nuclei Are Dustier than Their Hosts*, ApJ, 986, 177
56. Kocevski, D. D., et al. 2025, *The Rise of Faint, Red Active Galactic Nuclei at $z > 4$: A Sample of Little Red Dots in the JWST Extragalactic Legacy Fields*, ApJ, 986, 126
55. Wang, B., et al. 2025, *RUBIES: JWST/NIRSpec Confirmation of an Infrared-luminous, Broad-line Little Red Dot with an Ionized Outflow*, ApJ, 984, 121
54. de Graaff, A., et al. 2025, *RUBIES: a complete census of the bright and red distant Universe with JWST/NIRSpec*, A&A, 697, A189,
53. Finkelstein, S. L., et al. 2025, *The Cosmic Evolution Early Release Science Survey (CEERS)*, ApJL, 983, L4
52. Weibel, A., et al. 2025, *RUBIES Reveals a Massive Quiescent Galaxy at $z=7.3$* , ApJ, 983, 11

51. Cooper, O. R., et al. 2025, *RUBIES: JWST/NIRSpec resolves evolutionary phases of dusty star-forming galaxies at $z \sim 2$* , ApJ, 982, 125
50. de Graaff, A., et al. 2025, *Efficient formation of a massive quiescent galaxy at redshift 4.9*, Nature Astronomy 9, 280
49. Shen, L., et al. 2025, *NGDEEP: The Star Formation and Ionization Properties of Galaxies at $1.7 < z < 3.4$* , ApJL, 980, L45
48. Cole, J. W., et al. 2025, *CEERS: Increasing Scatter along the Star-forming Main Sequence Indicates Early Galaxies Form in Bursts*, ApJ, 979, 193
47. Zavala, J., et al. 2025, *A luminous and young galaxy at $z = 12.33$ revealed by a JWST/MIRI detection of $H\alpha$ and $[O III]$* , Nature Astronomy, 9, 155
46. Cheng, Y., et al. 2025, *Unveiling the Dark Side of UV/Optical Bright Galaxies: Optically Thick Dust Absorption*, ApJ, 979, 71
45. Bisigello, L., et al. 2025, *Spectroscopic confirmation of a dust-obscured, metal-rich dwarf galaxy at $z \sim 5$* , A&A, 693, L18
44. Rose, C., et al. 2024, *CEERS Key Paper. IX. Identifying Galaxy Mergers in CEERS NIRCам Images Using Random Forests and Convolutional Neural Networks*, ApJL, 976, L8
43. Llerena, M., et al. 2024, *Physical properties of extreme emission-line galaxies at $z \sim 4 - 9$ from the JWST CEERS survey*, A&A, 691, A59
42. Gupta, A. R., et al. 2024, *Emission-Line Ratios and Ionization Conditions of CEERS Star-Forming Galaxies with JWST/NIRSpec*, Research Notes of the American Astronomical Society, 8, 266
41. Davis, K., et al. 2024, *A Census from JWST of Extreme Emission-line Galaxies Spanning the Epoch of Reionization in CEERS*, ApJ, 974, 42
40. Calabró, A., et al. 2024, *The evolution of the star formation rate and Σ_{SFR} of galaxies in cosmic morning ($4 < z < 10$)*, A&A, 690, A290
39. Chworowsky, K., et al. 2024, *Evidence for a Shallow Evolution in the Volume Densities of Massive Galaxies at $z = 4$ to 8 from CEERS*, AJ, 168, 113
38. Seillé, L.-M., et al. 2024, *Physical properties of strong $1 < z < 3$ Balmer and Paschen lines emitters observed with JWST*, A&A, 689, A102
37. Hu, W., et al. 2024, *Characterizing the Average Interstellar Medium Conditions of Galaxies at $z \sim 5.6-9$ with UV and Optical Nebular Lines*, ApJ, 971, 21
36. Napolitano, L., et al. 2024, *'Peering into cosmic reionization: the $Ly\alpha$ visibility evolution from galaxies at $z = 4.5-8.5$ with JWST*, A&A, 688, A106
35. Ronayne, K., et al. 2024, *CEERS: $7.7 \mu m$ PAH Star Formation Rate Calibration with JWST MIRI*, ApJ, 970, 61
34. Wang, B., et al. 2024, *RUBIES: Evolved Stellar Populations with Extended Formation Histories at $z \sim 7 - 8$ in Candidate Massive Galaxies Identified with JWST/NIRSpec*, ApJL, 969, L13
33. Finkelstein, S. L., et al. 2024, *The Complete CEERS Early Universe Galaxy Sample: A Surprisingly Slow Evolution of the Space Density of Bright Galaxies at $z \sim 8.5 - 14.5$* , ApJL, 969, L2
32. Pirzkal, N., et al. 2024, *The Next Generation Deep Extragalactic Exploratory Public Near-Infrared Slitless Survey Epoch 1 (NGDEEP-NISS1): Extra-Galactic Star-formation and Active Galactic Nuclei at $0.5 < z < 3.6$* , ApJ, 969, 90
31. Jung, I., et al. 2024, *CEERS: Diversity of Lyman-Alpha Emitters during the Epoch of Reionization*, ApJ, 967, 73
30. Mascia, S. et al. 2024 *New insight on the nature of cosmic reionizers from the CEERS survey* A&A, 685, A3

29. Morales, A. M., et al. 2024, *Rest-Frame UV Colors for Faint Galaxies at $z \sim 9 - 16$ with the JWST NGDEEP Survey*, ApJL, 964, L24
28. Cheng, Y., et al. 2024, *Exploring the Gas-Phase Metallicity Gradients of Star-forming Galaxies at Cosmic Noon*, ApJ, 964, 94
27. Shen, L., et al. 2024, *NGDEEP Epoch 1: Spatially Resolved $H\alpha$ Observations of Disk and Bulge Growth in Star-Forming Galaxies at $z \sim 0.6-2.2$ from JWST NIRISS Slitless Spectroscopy*, ApJL, 963, L49
26. Barro, G., et al. 2024, *Extremely Red Galaxies at $z = 5-9$ with MIRI and NIRSpec: Dusty Galaxies or Obscured Active Galactic Nuclei?*, ApJ, 963, 128
25. Backhaus, B. E., et al. 2024, *CEERS Key Paper. VIII. Emission-line Ratios from NIRSpec and NIRCам Wide-Field Slitless Spectroscopy at $z > 2$* , ApJ, 962, 195
24. Kirkpatrick, A., et al. 2023, *CEERS Key Paper VII: JWST/MIRI Reveals a Faint Population of Galaxies at Cosmic Noon Unseen by Spitzer*, ApJL, 959, L7
23. Calabró, A., et al. 2023, *Near-infrared emission line diagnostics for AGN from the local Universe to redshift 3*, A&A, 679, A80
22. Fujimoto, S., et al. 2023, *ALMA FIR View of Ultra High-redshift Galaxy Candidates at $z \sim 11-17$: Blue Monsters or Low- z Red Interlopers?*, ApJ, 955, 130
21. Kocevski, D. D., et al. 2023, *Hidden Little Monsters: Spectroscopic Identification of Low-Mass, Broad-Line AGN at $z > 5$ with CEERS*, ApJL, 954, L4
20. Arrabal Haro, P., et al. 2023, *Spectroscopic confirmation of CEERS NIRCам-selected galaxies at $z \simeq 8 - 10$* , ApJL, 951, L22
19. Estrada-Carpenter, V., et al. 2023, *CLEAR: The Morphological Evolution of Galaxies in the Green Valley*, ApJ, 951, 115
18. Yang, G., et al. 2023, *CEERS Key Paper VI: JWST/MIRI Uncovers a Large Population of Obscured AGN at High Redshifts*, ApJL, 950, L5
17. Papovich, C., et al. 2023, *CEERS Key Paper IV: Galaxies at $4 < z < 9$ are Bluer than They Appear – Characterizing Galaxy Stellar Populations from Rest-Frame ~ 1 micron Imaging*, ApJL, 949, L18
16. Simons, R. C., et al. 2023, *CLEAR: Survey Overview, Data Analysis and Products*, ApJS, 266, 13
15. Constantin, L. et al. 2023, *Expectations of the size evolution of massive galaxies at $3 \leq z \leq 6$ from the TNG50 simulation: the CEERS/JWST view*, ApJ, 946, 71
14. Perez-Gonzalez, P. G., et al. 2023, *CEERS Key Paper V: A triality on the nature of HST-dark galaxies*, ApJL, 946, L16
13. Kocevski, D. D., et al. 2023, *CEERS Key Paper II: The Resolved Host Properties of AGN at $3 < z < 5$ with JWST*, ApJL, 946, L14
12. Finkelstein, S. L., et al. 2023, *CEERS Key Paper I: An Early Look into the First 500 Myr of Galaxy Formation with JWST*, ApJL, 946, L13
11. Guo, Y. et al. 2023, *First Look at $z > 1$ Bars in the Rest-Frame Near-Infrared with JWST Early CEERS Imaging*, ApJL, 945, L10
10. Trump, J. R. et al. 2023, *The Physical Conditions of Emission-Line Galaxies at Cosmic Dawn from JWST/NIRSpec Spectroscopy in the SMACS 0723 Early Release Observations*, ApJ, 945, 35
9. García-Argumánez, A. et al. 2023, *Probing the earliest phases in the formation of massive galaxies with simulated HST+JWST imaging data from Illustris*, ApJ, 944, 3
8. Zavala, J. et al. 2023, *Dusty starbursts masquerading as ultra high redshift galaxies in JWST observations*, ApJL, 943, L9

7. Rose, C. et al. 2023, *Identifying Galaxy Mergers in Simulated CEERS NIRCам Images using Random Forests*, ApJ, 942, 54
6. Finkelstein, S. L. et al. 2022, *A Long Time Ago in a Galaxy Far, Far Away: A Candidate $z \sim 14$ Galaxy in Early JWST CEERS Imaging*, ApJL, 940, L55
5. Papovich, C. et al. 2022, *CLEAR: The Ionization and Chemical-Enrichment Properties of Galaxies at $1.1 < z < 2.3$* , ApJ, 937, 22
4. Matharu, J. et al. 2022, *CLEAR: The Evolution of Spatially Resolved Star Formation in Galaxies between $0.5 \leq z \leq 1.7$ using $H\alpha$ Emission Line Maps*, ApJ, 937, 16
3. Jung, I. et al. 2022, *CLEAR: Boosted $Ly\alpha$ Transmission of the Intergalactic Medium in UV bright Galaxies*, ApJ, 933, 87
2. Simons, R. C. et al. 2021, *CLEAR: The Gas-Phase Metallicity Gradients of Star-Forming Galaxies at $0.6 < z < 2.6$* , ApJ, 923, 203
1. Estrada-Carpenter, V. et al. 2020, *CLEAR II: Evidence for Early Formation of the Most Compact Quiescent Galaxies at High Redshift*, ApJ, 880, 2

Co-Author: Submitted

15. Khullar, G., et al. 2026, *LEGGOS I: The JWST LEGGOS Survey – LEnsing and Galaxy Growth: Observing Substructures – Unpacks the Nature of Clumpy Star Formation and Quenching in Gravitationally Lensed Galaxies beyond Cosmic Noon*, arXiv e-prints, arXiv:2606.20845
14. Gentile, F., et al. 2026, *The quasi-star model for Little Red Dots: potential and challenges*, arXiv e-prints, arXiv:2606.06575
13. Davis, K., et al. 2026, *OCEANS of Absorption: High-resolution NIRSpectroscopy Reveals Diverse Balmer-line Absorption in Little Red Dots*, arXiv e-prints, arXiv:2606.00258
12. Hu W., et al. 2026, *Using $Ly\alpha$ Transmitted Spectrum to Probe IGM Transmission and Identify Ionized Structures in Cosmic Reionization*, arXiv e-prints, arXiv:2605.02539
11. Lambrides, E., et al. 2026, *The Glimmlr: Spectroscopic Variability in a $z \sim 7$ LRD Indicates Rapid Changes in Both the Narrow and Broad Line Regions*, arXiv e-prints, arXiv:2604.25991
10. Davis, K., et al. 2026, *Extreme Emission Line Galaxies in CEERS Are Powered by Star Formation, not AGN*, arXiv e-prints, arXiv:2602.23310
9. Wang, B., et al. 2026, *Water absorption confirms cool atmospheres in two little red dots*, arXiv e-prints, arXiv:2602.06024
8. Hutchison, T. A., et al. 2025, *THRILs – The High-(Redshift+Ionization) Line Search: Program Description & Redshift Catalog*, arXiv e-prints, arXiv:2512.12509
7. Lewis, Z., et al. 2025, *The Mass-Metallicity Relation and its Observational Effects at $z \sim 3-6$* , arXiv e-prints, arXiv:2512.03134
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3. Jones, B. L., et al. 2025, *The $M_{BH} - M_*$ Relationship at $3 < z < 7$: Big Black Holes in Little Red Dots*, arXiv e-prints, arXiv:2510.07376

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1. Jung, I, et al. 2022, *New $z > 7$ Lyman-alpha Emitters in EGS: Evidence of an Extended Ionized Structure at $z \sim 7.7$* , arXiv e-prints, arXiv:2212.09850

Presentations

Research Presentations (**bold** = invited)

23. **Talk:** *The Rest-Frame Optical Spectra of High-Redshift Galaxies: What are we seeing?*, at the California Institute of Technology, virtual 8 June 2026
22. **Talk:** *The Rest-Frame Optical Spectra of High-Redshift Galaxies: What are we seeing?*, at the Dartmouth College, Hanover, New Hampshire, USA 1 May 2026
21. **Talk:** *Characterizing High-Redshift Sources with Optical Emission Line Ratios*, at the Space Telescope Science Institute, Baltimore, Maryland, USA 14 August 2025
20. **Talk:** *Characterizing High-Redshift Objects with Optical Strong Line Ratios*, Winter Harbor, Maine, USA 28 July 2025
19. **Talk:** *A Census of Optical Diagnostics of Ionizing Sources Across Cosmic Time*, Bergen, Netherlands 8 May 2025
18. **Talk:** *High-Redshift Diagnostics of Star Formation and Active Galactic Nuclei*, The Pennsylvania State University, State College, Pennsylvania, USA 10 February 2025
17. **Talk:** *High-Redshift Diagnostics of Star Formation and Active Galactic Nuclei*, The Pennsylvania State University, State College, Pennsylvania, USA 17 September 2024
16. **Talk:** *High-Redshift Diagnostics of Ionizing Sources*, University of Wisconsin, Madison, Wisconsin, USA 17 July 2024
15. **Discussion Chair:** *Active Galactic Nuclei and Little Red Dots*, San Lorenzo de El Escorial, Spain 13 May 2024
14. **Talk:** *Diagnostics of Ionizing Sources at High-z using Models and Observations*, San Lorenzo de El Escorial, Spain 13 May 2024
13. **Talk:** *Diagnostics of AGN, Black Hole Seeds, and Population III Stars with JWST* at the AAS 243rd Meeting, New Orleans, Louisiana, USA 10 January 2024
12. **Poster:** *Emission Line Ratio Diagnostics of AGN, Black Hole Seeds and Population III Stars with JWST* at the First Year of JWST Science Conference, Space Telescope Science Institute, Baltimore, Maryland, USA 11 September 2023
11. **Talk:** *Diagnostics of Exotic Ionizing Sources with JWST* at Texas A&M Astrosymposium, College Station, Texas, USA 17 August 2023
10. **Talk:** *Diagnostics of Exotic Ionizing Sources Across Cosmic Time - High-Ionization Emission-Line Ratios: Ne53* at University of Texas, Austin, Texas, USA 10 May 2023
9. **Poster:** *High-Ionization [Ne V] Emission-Line Galaxies at Cosmic Noon and the Epoch of Reionization* at AAS 241st Meeting, Seattle, Washington, USA 12 January 2023
8. **Talk:** *Using [Ne V] to Constrain the Sources of Highly-Energetic Photoionization Across Cosmic Time: Exploring the "Mystery of Neon" with HST and JWST* at Texas A&M University, College Station, Texas, USA 2 December 2022
7. **Talk:** *Extreme High-Ionization Emission-Line Galaxies at Cosmic Noon and the Epoch of Reionization: Exploring the "Mystery of Neon" with HST and JWST* at Texas A&M University, College Station, Texas, USA 18 August 2022
6. **Talk:** *The Evolution of Spectroscopy from HST to JWST: Implications for the Epoch of Reionization* at Texas A&M University, College Station, Texas, USA 22 July 2022

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| | Poster: <i>HST Grism Observations of Paschen-Line Star-Formation and Dust Attenuation: A Precursor to the JWST Era</i> at AAS 240th Meeting, Pasadena, California, USA | 14 June 2022 |
| 5. | | |
| 4. | Talk: <i>Paschen-β Star Formation Rates and Dust Attenuation with HST and JWST</i> at Texas A&M Astrosymposium, College Station, Texas, USA | 27 August 2021 |
| 3. | Poster: <i>CLEAR: Paschen-β Star Formation Rates and Dust Attenuation in Low Redshift Galaxies</i> at AAS 237th Meeting, Virtual | 13 January 2021 |
| 2. | Poster: <i>Modeling 8B Solar Neutrino Detection with $CE\nu NS$</i> at AAS 233rd Meeting, Seattle, Washington, USA | 9 January 2019 |
| 1. | Poster: <i>Modeling 8B Solar Neutrino Detection with $CE\nu NS$</i> at TAMU Undergraduate Research Poster Session, College Station, Texas, USA | 1 August 2018 |

Outreach Presentations

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| 7. | "Broken Lecture": <i>Did JWST "Break The Universe"?</i> , State College, Pennsylvania, USA | 29 April 2026 |
| 6. | Talk: <i>Did JWST "Break The Universe"?</i> , State College, Pennsylvania, USA | 11 April 2026 |
| 5. | Talk: <i>Island Universes: Is the Milky Way Everything?</i> at Astronomy on Tap State College, State College, Pennsylvania, USA | 19 March 2026 |
| 4. | Talk: <i>Astronomy and You: The Impacts of Astronomy on Everyday Life</i> at Astronomy on Tap State College, State College, Pennsylvania, USA | 20 March 2025 |
| 3. | Talk: <i>The Evolving Universe Through JWST's Eyes</i> at the Pennsylvania State University, State College, Pennsylvania | 1 February 2025 |
| 2. | Talk: <i>The Origin of the Elements: Chemistry Across Cosmic Time</i> at Astronomy on Tap State College, State College, Pennsylvania, USA | 19 September 2024 |
| 1. | Talk: <i>Beyond the Telescope: Unraveling Mysteries with AI in Astronomy</i> at Astronomy on Tap B/CS, Bryan, Texas, USA | 24 April 2024 |

Professional Development Presentations

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| 10. | Panel: <i>Applying to Postdocs</i> at The Pennsylvania State University, State College, Pennsylvania, USA | 7 April 2026 |
| 9. | Panel: <i>Applying to Postdocs</i> at The Pennsylvania State University, State College, Pennsylvania, USA | 24 March 2025 |
| 8. | Panel: <i>GLASS Postdoc Panel</i> at Texas A&M University, College Station, Texas, USA | 8 March 2024 |
| 7. | Talk: <i>How to Be A Referee</i> at Texas A&M University, College Station, Texas, USA | 10 November 2023 |
| 6. | Panel: <i>How to Get Into Grad School</i> at Texas A&M University, College Station, Texas, USA | 28 July 2023 |
| 5. | Talk: <i>Data Visualization in Astronomy: More Important than the Science Itself?</i> at Texas A&M University, College Station, Texas, USA | 11 November 2022 |
| 4. | Panel: <i>How to Get Into Grad School</i> at Texas A&M University, College Station, Texas, USA | 29 July 2022 |
| 3. | <i>Data Visualization in Astronomy: More Important than the Science Itself?</i> at Texas A&M University, College Station, Texas, USA | 2 June 2022 |

2. Workshop: *Matplotlib: The Champion of Plotting in Python* at Texas A&M University, College Station, Texas, USA 2 June 2022
1. Workshop: *pandas: Your Best Friend for Data Analysis in Python* at Texas A&M University, College Station, Texas, USA 1 June 2022

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